

Test 26: PBC torus Dirac string

Verdict: PASS | Generated: 2026-09-15 03:07:01 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

Mixed: flux identities exact; ϵ on 4 neutral vortices over the bulk window (statistical).

METHOD: Mixed: flux identities exact; ϵ on 4 neutral vortices over the bulk window (statistical).

Test 26 HARDCORE pass 2: 96x96 torus, exhaustive artifact classification, n=5 ϵ realizations at $W_{\text{eff}}=1.00$.

Test 26 HARDCORE: artifacts 9024 clean/0 wrap/0 unexplained; $\epsilon=0.5964\text{--}0.0021$, scatter $\epsilon=0.0046$.

Test 26b [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

Plots

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 26 --no-two-pass`

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

[02:35:52.402] [+10143.430s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\epsilon=0.0000$, t=1.00, W=4.00, torus, 1 vortices, model=:dirac

[02:36:04.354] [+10155.382s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\epsilon=0.5000$, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

FULL COMPUTATION LOG

[02:41:24.785] [+10475.813s] calc: $\epsilon = 0.600632$ (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

[02:41:31.414] [+10482.443s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\epsilon=0.5000$, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[02:47:55.044] [+10866.072s] calc: $\epsilon = 0.596193$ (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

[02:47:58.346] [+10869.375s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\epsilon=0.5000$, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[02:54:12.143] [+11243.171s] calc: $\epsilon = 0.600551$ (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

[02:54:15.944] [+11246.972s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\epsilon=0.5000$, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[03:00:22.789] [+11613.818s] calc: $\epsilon = 0.589380$ (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

[03:00:28.175] [+11619.203s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\epsilon=0.5000$, t=1.00, W=1.00, torus, 4 vortices, model=:monumental

[03:06:55.993] [+12007.021s] calc: $\epsilon = 0.595447$ (n_r=5527, n_eigs=5529) [refs: GUE=0.5996 Poisson=0.3863]

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TEST 26 (pass 2, HARDCORE): PBC torus Dirac string ? deep validation

Exhaustive artifact classification + ϵ ensemble (n ? 5)

CONSOLE CAPTURE

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? RUNTIME: flux scan (1 build) + n=5 ϵ realizations at 96x96 ? expect ? 12?17 min

?r? row: ?=0.5000, Nv=4 (base protocol), q=1.0000, W=1.00 = ab_w_eff (capped)

H1 flux: vortex plaquette OK; empty plaquettes: 9024 clean, 0 wrap artifacts (0.0000), 191 seam (gauge holonomy, ix=96/iy=96), 0 unexplained

real 1/5: ?r? = 0.6006

real 2/5: ?r? = 0.5962

real 3/5: ?r? = 0.6006

real 4/5: ?r? = 0.5894

real 5/5: ?r? = 0.5954

Ensemble: ?r? = 0.5964 – 0.0021 | bootstrap CI [0.5928, 0.5997] ? GUE | above/below: 2/3

??

flux + artifacts PASS ? vortex OK; 9024 clean + 0 wrap(?2?q) + 191 seam holonomy; 0 unexplained

?r? > 0.5 (ensemble) PASS ? ?r? = 0.5964 – 0.0021 (n=5, base contract)

scatter ? 0.022 PASS ? ?=0.0046 ? ? monograph band

hermiticity ?real PASS ? verify_hermitian per realization

Test 26b [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS